



UNIVERSITY OF THE PHILIPPINES

Bachelor of Science in Physics

Your Name

Thesis Title

Thesis Adviser:

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This is to certify that this undergraduate thesis entitled **Thesis Title**, prepared and submitted by Your Name, is hereby endorsed for acceptance in partial fulfillment of the requirements for the degree of Bachelor of Science in Physics.

ADVISER NAME, Ph.D.
Thesis Adviser

This undergraduate thesis is hereby officially accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Physics.

WILSON O. GARCIA, Ph.D.
Director
National Institute of Physics

To a person/persons/idea/thing/etc...
This can be removed by commenting the following line:
`\dedication{\input{dedication.tex}}` in `main.tex`

Acknowledgments

I would like to acknowledge the help of the following..

My Adviser Name for whatever..

Others for whatever..

A team — First Member, Second Member, and Third Significant Member — for
whatever..

Thank you for all the support... blah blah blah.

ABSTRACT

THESIS TITLE

Your Name

University of the Philippines (2021)

Adviser:

Adviser Name, Ph.D.

This is where your abstract is!!!1

PACS: 01.20.+x [Communication forms and techniques (written, oral, electronic, etc.)], 01.30mm (Textbooks for graduates and researchers)

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Chapter 1

Introduction

1.1 Background of Study

1.2 Statement of the Problem

1.3 Objectives of the Study

The study aims to investigate the effects of coherence in the performance of the phase retrieval algorithm specifically, the Single-Beam Multiple-Intensity Reconstruction (SBMIR). Through understanding the principles of speckle contrast and formation, and coherence, a model for the use of partially coherent illumination in SBMIR will be developed. Phase, as well as the amplitude, reconstructions will be analyzed throughout the investigation. The model will utilize a recently proposed enhanced SBMIR method that involves non-sequential propagation which addresses some issues of the conventional one.

1.4 Significance of the Study

For years, partially coherent beams (PCB) have gained considerable attention because of their useful applications on different fields of optics and photonics. Compared to highly coherent beams, illumination from a partially coherent source reduces speckle effects and improves the image quality [?]. These partially coherent light

sources are found to be applied on free-space optical communications [? ?]. Implementation using a partially coherent light source could extend the utilization and application of phase retrieval.

By introducing PCB light sources to the phase retrieval algorithm, specifically the intensity recording part, the configuration cost can be reduced significantly since they are more insensitive to external perturbations. The ability to maximize the available intensity measurements and information will make the proposed technique more cost-effective. Moreover, numerical simulations can provide valuable insights on a viable model for phase retrieval using partially coherent light.

1.5 Scope and Limitation of the Study

The reconstruction algorithms to be analyzed in this study are applied on numerically simulated intensity images. Reconstructions from experimental data are not in the scope of the study. In addition, the main basis for the OMW method is the Fresnel diffraction integral which was consequently used to simulate wave propagation. However, the Fresnel diffraction integral is known to scale with propagation distance and wavelength which limits the parameters that can be used in the numerical simulations [6]. Lastly, phase retrieval methods involving multiple intensity measurements are highly dependent on the intensity variations. Consequently, complete wave reconstruction cannot be always guaranteed as the reconstruction algorithm is object dependent as well.

Chapter 2

Coherence Phenomenon

Light sources play significant roles on various optical processes and systems. Lights from known sources like lasers are nearly monochromatic and are known to be relatively coherent, while light from the sun is incoherent. The generated light beams from light sources of various degrees of coherence exhibit behaviors that are very different from one another.

Coherence refers to the amount of correlation in the optical field at separate times or separate points within a beam of light [?]. High coherence leads to stationary interference effects such as the “fringing” and “ringing” structures that appeared in the simulated diffraction and imaging results for monochromatic (coherent) light [?].

To be able to discuss coherence, it is convenient to first discuss classify coherence into two: temporal and spatial coherence.

2.1 Interference

- 2.2 Spatial Coherence
- 2.3 Temporal Coherence
- 2.4 Coherent Light
- 2.5 Incoherent Light
- 2.6 Partially-coherent Light

Chapter 3

Phase Retrieval

This is a customized chapter.

3.1 First section

You may change the filename of this file as long as you correspondingly change the filename stated in the `input{}` line in the `main.tex`.

3.2 Including figures

Figures can be added into your $\text{\LaTeX}$ files using the `figure` environment. However, it is **recommended** that you use `.eps` file format. These are encapsulated postscript files. This can be easily done by installing a postscript printer that outputs to a file (port is `FILE`). Ask your system administrator to install such a device for producing `eps` files by simply printing it to that device.

Shown in Fig. 3.1 is a figure created from Excel®. Note that the bounding box and page bounding box should be adjusted well enough to show the correct field.

Please see the sample chapter on how the figure is included in the `tex` files.

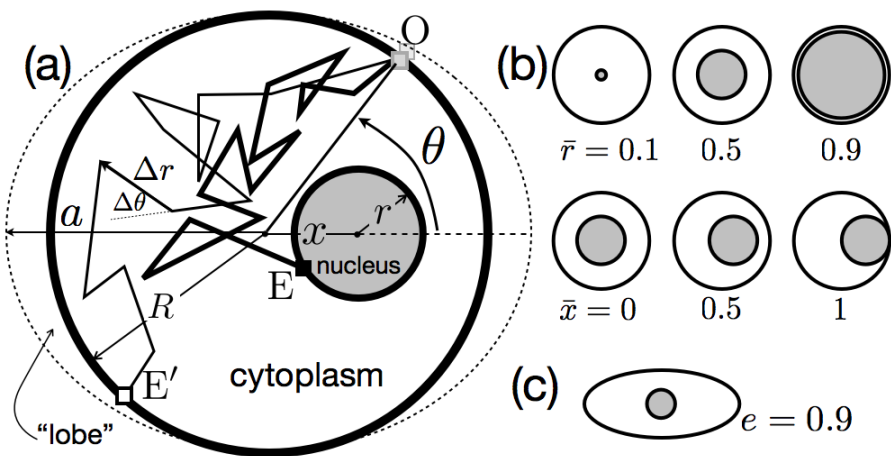


Figure 3.1: A *.png file.

Chapter 4

Results and Discussion

In this chapter, a sample for making equations and sub-equations are demonstrated.

4.1 Equations and sub-equations

In the following, a set of equations is shown.

$$\vec{\nabla} \cdot \vec{D} = \rho \tag{4.1a}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \tag{4.1b}$$

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B} \tag{4.1c}$$

The last one being

$$\vec{\nabla} \times \vec{H} = \vec{J} + \partial_t \vec{D} \tag{4.1d}$$

Note that text can still be placed between sub-equations within the `subequations` environment.

When using a solitary equations, you may use the usual equation syntax in L^AT_EX.

$$E = mc^2 \tag{4.2}$$

Chapter 5

Summary and Conclusions

A short sample thesis/dissertation is presented. Although not complete, it will be useful for newbies in L^AT_EX. Any questions? email me at the following address: johnrob.bantang@gmail.com.

Appendix A

Sample appendix

This gives an example of an appendix chapter. Note that this file has been included **after** the line `\appendix` in `main.tex`.

A.1 Equations in appendix

You don't need to worry about equations within the appendix since $\text{\LaTeX}$ automatically formats the equation numbers for you. For example,

$$c^2 = a^2 + b^2 \tag{A.1}$$

becomes the Pythagorean theorem where c is the length of the longest side of any right triangle.

A.2 Codes as appendix

Include your codes when necessary to your thesis/dissertation. To do this, you may use `verbatim` environment as follows. **WARNING:** All `verbatim` and `verbatiminput` environments should always be treated as a separate paragraph. When included in a text paragraph, it sometimes happen to reduce the 1.5 spacing to the usual single-spaced text.

```
#include <iostream>
using std::cout;
using std::endl;

int main( void )
{
    cout << "Hello world!" << endl;
    return 0;
}
```

The `{\small }` bracketed region is used to lower the font size of the entire verbatim text. This will save you much space and give a more aesthetical look in your manuscript.

On the other hand, when very long codes are wished to be included automatically without the tedious cut and paste procedure, you may include them using the `\verbatiminput` command as follows. You may want to include a short description of the code of course.

```
//Johnrob Y. Bantang, Natinal Institute of Physics
//Created: 03 October 2002
// Makes new C files
// usage: newC filename
//Modifications:
// >> 21 Jan 2003, Johnrob
//   included the constant AUTHOR and AFFILIATION for portability

#include <stdlib.h>
#include <iostream.h>
#include <fstream.h>
#include <strstream.h>
#include <time.h>
#include <string.h>

const char *const AUTHOR= "Johnrob Y. Bantang";
const char *const AFFILIATION= "National Institute of Physics";
const char *const EXTENSION= ".cpp";

int main(int argc,char **argv){
if(argc!=2){
cout<<"usage: newC filename"<<endl;
exit(0);
}
char *fname= new char[strlen(argv[1])+5];
ostrstream out(fname,strlen(argv[1])+5);
out<<argv[1]<<EXTENSION;
time_t date=time(NULL);

ofstream file(fname,ios::nocreate,0);
//opens normal file that **already exists**;
if(file){
for(int i=1;i<5;i++)
cout<<"WARNING! file aready exists!"<<endl;
cout<<endl<<"you can type"<<endl<<endl;
cout<<"\t\"head "<<fname<<"\"<<endl;
cout<<"in command line to *view version*"<<endl<<endl;
cout<<"please enter 1 to OVERWRITE this file"<<endl;
```

```

cout<<"type anything to cancel"<<endl;
for(int i=1;i<5;i++)
cout<<"WARNING! file already exists!"<<endl;
int n;
cin>>n;
if(n!=1){
cout<<"*no* file is created... exiting..."<<endl<<endl;
exit(0);
}
file.close();
file.open(fname);
if(!file)
cout<<"**cannot create new file!**"<<endl;
cout<<"OLD FILE: "<<fname<<" *overwritten!*"<<endl;
}
if(!file){
file.open(fname);
if(!file){
cout<<"**cannot create new file!**"<<endl;
exit(0);
}
cout<<endl<<"NEW FILE created: "<<fname<<endl<<endl;
}

cout<<"\tcreating contents for the new C++ file: "<<endl<<endl;
//creating headers...
file<<"//filename: \""<<fname<<"\"<<endl;
file<<"//\"<<AUTHOR<<\", \"<<AFFILIATION<<endl;
file<<"//Created: \"<<asctime( localtime(&date) );
//writes the time and date today; endl already in asctime();
file<<"//\"<<endl;
file<<"//Comments:\"<<endl;
file<<"// >>\"<<endl;
file<<"//\"<<endl;
file<<"//This file is generated using the \"newC generator\"...\"<<endl;
file<<"//Modifications:\"<<endl;
file<<"// >>\"<<endl<<endl;
file<<"#include <iostream.h>\"<<endl;
file<<"#include <math.h>\"<<endl<<endl;
//starting the main body...
file<<"int main(int argc,char **argv){\"<<endl;
file<<"//\tif(argc!= ...){\"<<endl;
file<<"//\t\tcout<<\"usage: \"<<argv[1]<<\".exe ... \"<<endl;<<endl;
file<<"//\t\texit(1);\"<<endl;
file<<"//\t}\"<<endl;
file<<"\t//write the main body here\"<<endl;
file<<"return(0);\"<<endl<<\"}\"<<endl;

delete fname;

```

```
    file.close();  
    cout<<endl<<"\tCREATION SUCCESSFUL!"<<endl<<endl;  
return 0;  
}
```

This time, you may just include your recent codes by just copy-paste-ing the codes (as long they are clean!) into the directory `codes/` in the directory where this file is saved.

Bibliography

- [1] G.A. Mourou, C.P.J. Barty, and M.D. Perry. Ultrahigh-intensity lasers: Physics of the extreme on a tabletop. *Applied Optics*, 51(1):22–28, January 1998.
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